5 A loudspeaker produces a sound wave in air. The distance between the centre of a compression and the centre of a neighbouring rarefaction is 0.10 m.

(a) Calculate the wavelength of the sound wave.

wavelength = ... [1]

(b) State a typical value for the speed of sound in air.

..... [1]

(c) **(i)** Calculate the frequency of the sound from the loudspeaker.

frequency = ... [2]

(ii) Explain whether the sound from the loudspeaker is audible to a human with normal hearing.

.....

..... [1]

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(d) Another loudspeaker produces a sound of wavelength 0.40 m. Sound from the loudspeaker reaches a sound absorbing surface with a gap of width 0.80 m at the centre.

Fig. 5.1 shows the arrangement.

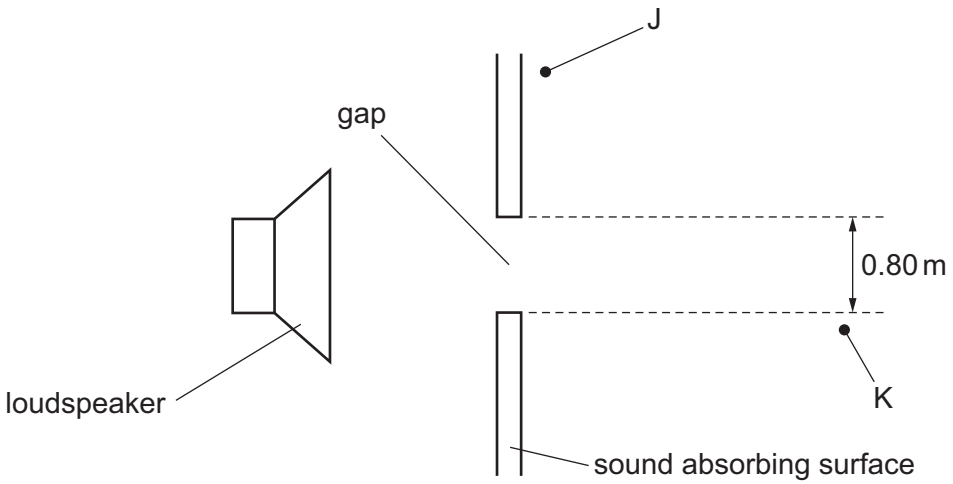

Fig. 5.1

Explain whether it is possible to detect sound from the loudspeaker at either point J or at point K.

point J .....

.....

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point K .....

.....

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[4]

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